

èfjç» - ç” µè`❖i¼š188-5665-3017 - é,®ç®±i¼šhs3434@foxmail.com -
ä,ªºç½ ‘ç«™i¼šhs3434.github.io - **æ±
,è❖Ææ,,❖â❖ ‘**i¼šAMD ROCm Python â❖Žç«`â¼â❖ ‘â • ¥ç`â, ^i¼^ROCm Radeon Cloud
æ-¹â❖ ‘i¼% --- ## ä,ªºç®€ä» 3 â¹´ç” Ýç%©äçjæ❖-â • ¥ç`âÆ-ç»❖éâEi¼Æè ±ªâ|è½-âžç
K8s â❖Žç«`æ ^â€, \*\*â®¹â™™âÆ-éf`ç½²â€❖èµ,,æº❖èºfâº|â€❖ä»»âŠç¼-æŽ’ â€❖GPU
æŽ`ç❖ + æœ❖âŠç¹\*\*â€, æœ€èç 2 â ‘`è ±ªâ| ROCm / K8s / Helm / FastAPI /
Prometheusi¼Æç«-ç«â®Ææ^❖ **k8s-llm-runtime**i¼^K8s ä, Šçš,, vLLM
æ`jâžçæœ❖âŠç½ ‘â...³i¼%oï¼Æ\*\*v0.1.0 taggedi¼Æâ❖`çŽºâœºè • ‘demo**â€, ç + Ýæ,%
Snakemake ç±» Argo Workflows çš,, ML â¹³â❖æŠ½è±jâ€, --- ## æ±,è❖ÆæŠ€èf½i¼^æÆ%â²
—â½❖âE¹é...❖æŽ’ âº❖i¼Æ4 â±,i¼% ### â❖Žç«`æ ,âçf - Python 3.11+i¼^asyncio, type
hints, Pydantic, httpx asynci¼% - FastAPIi¼^OpenAPI / dependency injection / lifespan¼% -
æµ«è` • i¼špytest, pytest-asyncio, 88% coverage ### â®¹â™™â, Žç¼-æŽ’ - Dockeri¼^multi-
stage æž,,âºi¼ÆAlpine é • œâf❖ä¼~âÆ-i¼% - Kubernetesi¼^Pod / Service / Deployment /
StatefulSet / ConfigMap / RBACi¼% - Helm charti¼^values æ`jæ❖;âÆ-i¼Æhelm install/
upgrade/uninstalli¼ÆCI lint cleani¼% - é • œâf❖ä» “âº “i¼šregistry.k8s.io / docker.io /
ghcr.io / nvcr.ioi¼^â½â + ... DaoCloud é • œâf❖âŠ é€Yi¼% ### GPU / LLM ç” Ýæ€❖ -
vLLMi¼^OpenAI â...¼â®¹ APIi¼ÆæŽ`ç❖ + æœ❖âŠçj¼% - AMD ROCm âºç;€i¼^HSA / HIP
æ|,âçµi¼Æamd.com/gpu èµ,,æº❖èºfâº|i¼% - PyTorch¼^CNN æ— ¶ç©ºè Š £ç ❖â™™ /
Transformeri¼% - Hugging Facei¼^transformers âº “i¼Ætokenizeri¼% ###
â❖`è Š ,æµ«æ€ Š ä, Žâ • ¥â... é “¾ - Prometheusi¼^prometheus_clienti¼Æhistogram /
counter / gaugei¼% - structlogi¼^JSON structured loggingi¼% - uv / ruff / mypy strict / GitHub
Actions - Linux æœ❖âŠçç«`èç❖ç»`i¼^systemd, Nginx, ç«`â❖£/èçç`ç®ç❖ + i¼% --- ##
éj¹ç®ç»»éâE ### k8s-llm-runtimei¼šK8s ä, Šçš,, vLLM æ`jâžçæœ❖âŠç½ ‘â...³i¼^v0.1.0
taggedi¼% **é—éç~**i¼^â-¹âº” JD è❖Æè`£ 1 + 3i¼%oï¼švLLM æ`jâžçéf`ç½²éœ€ GPU +
â®¹â™™ + è • `ç” ±i¼Æâçé~Ýâšäºâ...±âºæ— ¶ æ`jâžçç®ç❖ + âºæ❖, i¼ç” ç© ¶ /
âºâçé~Ýæf³ç” “ OpenAI â...¼â®¹ API æ— ¶ é—æ Š é«~â€, i¼^ç`æŽ¥â-¹âº”
JD i¼š"âš Š æ`jâžçèç»»f/æŽ`ç❖ + Workshopæ❖æ • ™ç`â€❖âŠÝèf½æ¼” ç«º”i¼%
**æž ¶ æž,, **i¼^â❖Æ chart + ä, %â±, lib + é • œâf❖ä¼~âÆ-i¼%oï¼š - `charts/llm-inference/
`i¼švLLM è`Ýè½½ charti¼Æ**GPU vendor â ± æ❖çä»...é❖ `values.yaml` **i¼^amd.com/
gpu / nvidia.com/gpu / none ä, %æ€❖i¼Ævalues æ~¾â¼â❖é€æºæ©i¼% - `charts/llm-router/
`i¼šFastAPI ç½ ‘â...³ chart - ConfigMap æ³`â...¥ initContainer âŠ è½½â...â¼Ý chart - RBAC
æœ€âºæ❖fè™i¼^Pod/Service/ConfigMap read+write + Leasei¼% - ServiceMonitor â❖`é€
%â❖`ç” “i¼^Service 0.0.0.0:9090/metricsi¼% - `src/k8s_llm_runtime/i¼š3 â±, lib æŠ½è±
j - `K8sJobOperator`i¼^â½Žâ±,i¼Æé€šç” “ Job èºfâº|i¼% - `VLLMInferenceOperator`i¼^â, -
â±,i¼Æhelm CLI subprocess + GPU vendori¼% - `ModelOperator`i¼^é«`â±,i¼ÆOpenAI â...
¼â®¹ + è ±ªâŠ`éf`ç½²i¼% **4 âš Š æŠ€æœ-âºç, **i¼š 1. **lib ä, %â±, â^ + â±, æŠ½è±
j**i¼^â-¹âº” JD â❖Žç«`æž ¶ æž,,èf½âŠi¼% - ä, Šâ±, ModelOperator â❖â...³âçf
load_model(alias) / unload_model(alias) - ä, â±, VLLMInferenceOperator â❖è£... helm
ç» + èŠ, - â½Žâ±, K8sJobOperator é€šç” “ K8s Job èºfâº| 2. **Helm ConfigMap æ³`â...¥
aliases**i¼^â-¹âº” JD â®¹â™™âÆ- + èµ,,æº❖èºfâº|i¼% - é❖çâ...❖æ`æ-æjæ-âçžæ`jâžçéf½
rebuild é • œâf❖i¼^GPU é • œâf❖ 15-20GB rebuild 5-10mini¼% - ConfigMap
ç Š` çº Š ç” Ýæ • ^i¼Æé ¶ é • œâf❖è ± ❖âº 3. **K8s Lease â^ + â, fâ¼â❖é” ❖**i¼^â-¹âº”
JD é + ç¾æèç❖ç»` + HAi¼% - coordination.k8s.io/v1 Lease é~²æç Router
âššâ% æœ-â¹ ¶ â❖ ‘éf`ç½²â❖Æä, €æ`jâžç - é” ❖ç²` âºæÆ% model
aliasi¼Æè ±ªâŠ`èç ± æœÝ + ç»çº| 4. **ä, šâŠçº Š Prometheus metrics**i¼^â-¹âº” JD
â¹³â❖â❖`è Š ,æµ«æ€ Š i¼% - `chat_completions_duration_seconds`i¼^histogrami¼Æâ❖«
status code æ ±ç¾âi¼% - `http\_requests\_total`i¼^counteri¼ÆæÆ% method/route/statusi¼%

models_loaded`i4^gaugei4Cæ` in-memory statei4% - ServiceMonitor â`ée%èè`
Grafana **GPU vendor â`æ` demo**i4^â`1^â` JD âŠ â` + GPU é` + ç34æi4%oi4%4š -
values.yaml`gpu.vendor=amd|nvidia|none` - AMDi4šamd.com/gpu èµ,,æ°è°fâ°i4CEROcm
vLLM é • œâf` - NVIDIAi4šnvidia.com/gpu èµ,,æ°è°fâ°i4CUDAvLLM é • œâf` -
nonei4šCPU only æ`jâ4^i4^kind cluster é`~è°æi4Cæœ-âœâ4€â` ‘ç’ `i4%` **è
`è è è`äèèè`**i4š - 31 commits / 90 tests pass / coverage 88% - ruff + mypy strict + helm-lint
â...` clean - 2 Helm chartsi4^llm-inference + llm-routeri4%` **è, ©â` ‘ä, Žäè°â`**i4š -
Service DNS ä, âC1é...` bugi4š`--set fullnameOverride={release_name}` äè°â` - kind
èŠ,ç,1â½â + ...é • œâf`âŠ ééYi4šcontainerdConfigPatches + DaoCloud mirror -
èèTMä°éf1½â + TMâ...¥` docs/amd-interview-demo.md` ä34çZ°âœâ`èèf
**æœâæ`\*\*i4^v1.1i4%oi4šSSE æµ`â4æZ`ç` + / RBAC æ` ¶ ç´ S â`è` » / çœYâ°Ž GPU
é` + ç34æ e2e ### âY°ä°Ž Snakemake çš,, ML
â13â`â • ¥â½œæµ`æj + æž ¶ i4^ç` YäèèfCæTM`i4Cçš» Argo Workflowsi4% -
èfCæTM`i4šSnakemake rule è`æ³ • æ— æ³ • â°žçZ°æ`jâ`—és`æ,,ç»,â`^ -
è°¾è°èji4šæ ‘â4f rule è`æ³ • i4Cç` æZ¥è°fâ + ...éf`æZ¥â`£âŠ`æèç` Yæ^`
rulei4Cè S,,èCèè¾ “â...¥è¾ “â`°ç±»âž`â`â3â`ç S`æœ`â4æœ`14æZ¥ - â...³è`” i4šJD
âŠ â` + éj1”æœ°âTM`âjâ1 â13â`°ç`” â` ‘ç»é°C”â€” â€” æœ-éj1ç°æ`~ ML â13â`æS½è±
jçš,,â • ¥ç`â°žè • µ ### è,, ‘æœ°æZ¥â`£ CNN è S £ç`âTM`i4^PyTorch i4%â€” â€” 6 â` “â -
jâ1 è • `ç°èä° S â`° - ç«`â°ç«`æ— ¶ ç©°â` • ç S`è S £ç` EEG äèjâ` • - PyQt â°žæ— ¶ GUI +
FastAPI æš´éœ² - è`è S` BCI è • `ç°èç°€âž + [github.com/hs3434/bci-pipeline-demo](https://
github.com/hs3434/bci-pipeline-demo) --- ## â • ¥â½œç»âž + ### ä, Šæµ • â°
æèè Cæâ`jâèjæ`ç S` ‘æŠæœ°éTM`â...-â` , i½œ ç` Yç%©äèjæ`â • ¥ç`â , ^ i½œ
2025.04 - è`è³â»Š - \*\*Docker éf`ç12**i4šä, šäŠjâ»£ç` Docker âCè-i4^âY°ä°Ž snakemake +
èè`ç`” æj + æž ¶ i4% - **æœ`âŠjç«`èèç»`\*\*i4šLinux æœ`âŠjâTM`æ—
¥â, , ç» æŠæi4C£Ngix â`â`â` ‘â»£ç` + i4Csystemd æœ`âŠjç°jç` + -
**â°` â13â`°â • ¥â... • â4€â` \*\*i4šä, šäŠjèè`âŠ`âCè-æµ`ç`ç` + ç°—æ³ • â • ¥â... • ###
ä, Šæµ • æ- S æ` “ç” Yç%©âC»âjç S` ‘æŠæœ°éTM`â...-â` , i½œ
ç` Yç%©äèjæ`ç`” â` ‘â • ¥ç`â , ^ i½œ 2022.08 - 2024.07 - \*\*â°` â13â`°ç” Yäèjâ • ¥â...
• â4€â` \*\*i4šR / Python / Linux / Docker / â°` âžYç” Y - **ä, šäŠjèè`âŠ`âCè-**i4šâCè»â -
jè½-â½ • ç»,ä, šäŠjæµ`ç`èè`âŠ`âCè-i4Cæ`jâ`—âCè-âè`â`â`â`ç”` -
â13â`°âœ°â`€i4š[cloud.oebiotech.com](https://cloud.oebiotech.com/#/home) ###
ä, Šæµ • æ- S æ` “ç” Yç%©âC»âjæœ°éTM`â...-â` , i½œ ç” Yç%©äèjæ`â°žâ1 ç” Y i½œ
2021.10 - 2022.01 - â`ç` + èfžè½-â½ • ç»,i4šè
`æž S æè`â , è S,,â` + æž`æè`â` °â`žæè`ä, æè S âCè-â` + æž` -
ä, »èèæŠæœ`i4šâ`ç» + èfžè½-â½ • ç»,âè`Râè`Pythonâè`Linux --- ##
æ • TMè,²èfCæTM` \*\*è¥èâCè—â + œæž—ç S` ‘æŠæâ S âj1** i½œ æµ`ç°©ç S` â -
jâ, ŽæŠæœ`â, “â, š i½œ æœ-ç S` ‘i½œ 2018.09 - 2022.07 i½œ GPAi4š3.35 - æ` • ä, šè°æ-
è i4šâèŠâY°ä°žéè¾ “æ-è” ,âžâ½` çj°â°šâ°é°jâ€” â4ç»,â...³æè S çš ¶ â`ŠæŠ—
â€” æè S çš ¶ æCèèèèèè - âµ S â`éj1ç°i4šæ1ç`mè`CæZ¥ç S`â`1ç`«èŠèèèèèè “èæŠ—
é “œæ±jæY “çš,,â½±â “i4^â`ä, Žâ½œèè...i4% --- ## ä, â°â14^âŠè - \*\*6 â` “èè`âj K8s
â`žç«`æ`\*\*i4šâ»žè` ¶ â` v0.1.0 tagged éj1ç°i4Cç«`ç«`â`Cæ`â`C Helm chart + Python
lib - **çZ°âœ° demo èf½âš**i4šâ`èè`â , jç” µè,, ‘âœ` kind cluster ä, Šè • ‘éèšâ`Cæ •
LLM æZ`ç` + æµ`ç`ç`i4^éf`ç12â +` è • `ç” ±â +` chat completions â +` metricsi4% -
âjâ1 èf½âšâ4°i4š3 â1`â + ...â»žç” Yç%©äèjæ`è • `ç • Cæ^° K8s
â`žç«`i4CæŠæœ`çfæf...é«` - **3 â1` Python â • ¥ç`âCè-
ç»é°C**i4šâ»žç S` ‘ç” è,,šæœ-â`â°` â13â`°â • ¥â... • çš,,â°Cæ • `è • `â¾,,